Digital Twin Challenge 2026 | Happiest Health Summit

GlucoTwin-India

Multimodal Physiological Digital Twin for Proactive T2DM Glycemic Forecasting

Presenters: Team GlucoTwin (Lead: Ishika Mehere)

G H Raisoni University, Amravati

India's Diabetes Crisis: The Cost of Reactive Care

101M+

DIAGNOSED PATIENTS

Diagnosed diabetes patients nationwide requiring continuous care management.

136M

PRE-DIABETIC ADULTS

High-risk pre-diabetic adults currently lacking proactive preventive protocols.

- **100% reactive standard care:** Current healthcare pathways focus exclusively on symptom management rather than early prevention.
- **Delayed action:** Medical intervention only occurs post-acute glucose spike, increasing long-term health risks and treatment costs.

Beyond Wearables: What Is a Healthcare Digital Twin?

DISCONNECTED DATA STREAMS

Traditional Limitations

EHR System Constraint

Static EHRs lack temporal context.

Wearable Sensor Constraint

Wearable CGMs lack baseline clinical depth.

INTEGRATED SOLUTION

The Digital Twin Advantage

Unified Intelligence

Digital Twin synchronizes both data streams.

Predictive Horizon

Forecasts metabolic outcomes 2 hours in advance.

Sandbox Compliance & DPDP Act 2023

Zero PHI Risk

100% compliant with India's DPDP Act 2023 and HIPAA.

+ Synthea-Modeled EHR Profiles

Comprehensive synthetic patient records with HbA1c ranging from 6.5% to 9.5%.

+ Wearable Telemetry Stream

5-minute sampling interval wearable telemetry (CGM, HRV, Steps).

End-to-End Multimodal System Architecture

STAGE 01

Dual-stream Ingestion

Continuous parallel acquisition of clinical data streams and real-time patient metrics.

STAGE 02

Branch Encoders

32D Dense EHR

64D 2-Layer LSTM

STAGE 03

Late Fusion Brain

Unified representation processing engine.

OUTPUT VECTOR

96D Concatenated

STAGE 04

Clinical Cockpit & Alerts

Real-time risk scoring, actionable clinical dashboards, and automated alert triggers.

Active Monitoring

Model Validation & Clinical Safety Metrics

91.4%

Validation Accuracy

98.6%

Predictions in Clarke Error Grid Zones A & B

0

Clinical Hazards

80 / 20

Patient-Stratified Train-Test Split

Clinical Cockpit: Operationalizing the Virtual Patient

TRI-ZONE RISK ALERTS

- **Euglycemic Zone:** Target stability maintained
- **Spike Alert:** Hyperglycemia threshold (>180 mg/dL)
- **Crash Alert:** Hypoglycemia threshold (<70 mg/dL)

Real-Time Telemetry Curve

Continuous stream modeling with integrated 2-hour forward lookahead to anticipate glucose fluctuations.

In Silico Intervention Testing

Simulate therapeutic adjustments and dosage impacts dynamically prior to clinical execution.

Operationalized Decision Support

Seamlessly bridges virtual patient predictive models into actionable bedside clinical workflows.

Counterfactual 'What-If' Physiological Modeling

UNMITIGATED SCENARIO

210 mg/dL

Acute Glucose Spike

90g carbohydrate meal projects an acute spike to 210 mg/dL.

PRESCRIPTIVE SCENARIO

162 mg/dL

Safe Attenuation Target

20-minute postprandial walk attenuates glucose to a safe 162 mg/dL.

Scalability & National Health Integration

Ayushman Bharat Digital Mission (ABDM)

Native compatibility with Ayushman Bharat Health Accounts (ABHA IDs).

- Lightweight edge inference (120KB PyTorch weights).
- Affordable integration with commodity wearable bands in Tier 2/3 cities.

Transforming Chronic Care Across Bharat

OUR VISION

"Zero preventable diabetic hospitalizations through predictive digital twins."

PROJECT MILESTONE

Working end-to-end prototype submitted.

UPCOMING SHOWCASE

Ready for November 2026 Summit Showcase in Bengaluru.